

ECO 362 Financial Investments

Precept 8: Life Cycle Theory and Term Structure

November 10 – 14, 2025

1 Lifetime Cycle Theory

2 Bonds: Term Structure Model

- Consider human capital a type of asset that generates income through labor
- Idiosyncratic risk (injury, death, unemployment) with expected income $\mathbb{E}[L]$ each year
- If you plan to work for T years, present value of human capital/labor income is:

$$H = \sum_{t=1}^T \frac{\mathbb{E}[L]}{(1 + Y_b(t))^t} = \mathbb{E}[L]P_a(T)$$

- As before, $Y_b(t)$ is the zero-coupon yield with maturity t , and $P_a(T)$ is a T year annuity paying one dollar per year.

- How to optimally insure human capital?
 - Buy annual renewable life insurance: pay P premium today, and the insurance company pays F if you die within the next year (probability q).
- Insurance is actuarially fair if

$$P = \frac{qF}{1 + R_b}$$

with riskless rate R_b .

- Price assumes insurance companies have perfect competition
- Intuitively: the price for one year of insurance must be equal to the expected payout of that insurance.

- Given the present value of lifetime labor income H . How much insurance F should the investor buy?
- Maximize mean-variance expected utility over future wealth:

$$\max_F \mathbb{E}[W] - \frac{\gamma}{2} \text{Var}(W)$$

Impact on expected wealth:

- Result 1 (Lecture 9 LCT p.9): insurance F does not affect expected wealth.
- Expected value of entering life insurance is zero.

Impact on variance of wealth:

- Thus, the only reason to buy insurance is to minimize variance.
- Result 2 (Lecture 9 LCT p.10): $F = H$ minimizes the variance of future wealth.
- Intuition: insurance guarantees receiving H regardless of if the investor dies.

Portfolio Return with Human Capital

- Investor has both financial wealth A and human capital H .
 - Normalized to $(A + H = 1)$
- Financial wealth: invest in bond/stock
 - Bond: riskless return $(1 + R_b)$
 - Stock: expected return μ and variance σ^2
- Human capital: earn return $1 + R_b$
- Portfolio return:

$$\begin{aligned}1 + R_p &= (A - S)(1 + R_b) + S(1 + R_s) + H(1 + R_b) \\&= (A + H - S)(1 + R_b) + S(1 + R_s) \\&= \left(\underbrace{\left(1 - \frac{S}{A + H}\right)}_{1-w} (1 + R_b) + \underbrace{\frac{S}{A + H}}_w (1 + R_s) \right) \underbrace{(A + H)}_{=1}\end{aligned}$$

Portfolio Choice with Human Capital

- Investor maximizes mean-variance expected utility.

$$\max_w \mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p)$$

- (Remember this formula for the optimal portfolio?) the optimal w is...

$$w = \frac{S}{A + H} = \frac{S}{A(1 + H/A)} = \frac{\mu - R_b}{\gamma \sigma^2}$$

- Therefore,

$$\frac{S}{A} = \left(1 + \frac{H}{A}\right) \frac{\mu - R_b}{\gamma \sigma^2}$$

- Intuition: Since human capital is like a non-tradable bond, the more of it you have, the more of your financial wealth you should invest into stocks.
- See "LCT Examples.xlsx".

1 Lifetime Cycle Theory

2 Bonds: Term Structure Model

- A term structure model is a model on zero-coupon yield $y_s(t)$ in year s and with t years to maturity.
- The model describes
 - the relation between zero-coupon yields with different maturities: $y_s(t_1)$ and $y_s(t_2)$
 - the evolution of zero-coupon yields over time: $y_{s_1}(t)$ and $y_{s_2}(t)$

Term Structure Model: Assumptions and Main Expression

- 1 Expectations hypothesis (EH): Expected excess returns are constant:

$$\mathbb{E}_s[r_{s+1}(t) - y_s(1)] = \mu, \forall s \geq 0, t > 1. \quad (1)$$

- 2 Short rates are mean reverting:

$$y_s(1) - \bar{y} = \rho(y_{s-1}(1) - \bar{y}) + \epsilon_s. \quad (2)$$

We will skip the derivation details here. Under these assumptions, the yield curve is given by:

$$y_s(t) - \bar{y} = \frac{1}{t} \frac{1 - \rho^t}{1 - \rho} (y_s(1) - \bar{y}) + \frac{t-1}{t} \mu, \quad (3)$$

$$y_s(t) - \bar{y} = \frac{1}{t} \frac{1 - \rho^t}{1 - \rho} (y_s(1) - \bar{y}) + \frac{t-1}{t} \mu,$$

- The model explains facts:
 - 1 Mean reversion: tendency to rise when low and fall when high.
Due to the mean reversion of the short rates.
 - 2 Level factor: Riskless rate of all maturities tend to move together.
When $y_s(1)$ moves, so do $y_s(t)$ for all t .
 - 3 Volatility: short-term rates are more volatile than long term rates.
The coefficient $\frac{1}{t} \frac{1 - \rho^t}{1 - \rho}$ is less than 1 and is decreasing in t .

$$y_s(t) - y_s(1) = \underbrace{\left(\frac{1}{t} \frac{1 - \rho^t}{1 - \rho} - 1 \right)}_{<0} (y_s(1) - \bar{y}) + \frac{t-1}{t} \mu,$$

- Slope depends on two effects
 - 1 Expectation of future short rates
 - If $y_s(1) > \bar{y}$, the short rate is expected to fall, making the yield curve more downward sloping.
 - If $y_s(1) < \bar{y}$, the short rate is expected to rise, making the yield curve more upward sloping.
 - 2 Term premium (usually positive) makes the yield curve more upward sloping.

- Expected returns in the coming year are the same for all bonds (with maturities longer than one year).
 - This is a result of the expectations hypothesis:

$$\mathbb{E}_s[r_{s+1}(t)] = y_s(1) + \mu, \forall t. \quad (4)$$

- A higher yield is offset by a lower expected capital gain:

$$\mathbb{E}_s[r_{s+1}(t)] = \underbrace{y_s(t)}_{\text{yield}} + \underbrace{(t-1)\mathbb{E}_s[y_s(t) - y_{s+1}(t-1)]}_{\text{capital gain}} \quad (5)$$